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SPACE SCIENCE WITH PYTHON

Space Science with Python — A very bright Opposition

Part 19 of the tutorial series dives deeply into asteroid research: How does the brightness of an asteroid change? What are the dependencies and are there any amazing effects?

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Image composition of the dwarf planet Ceres. Images from the mission Dawn have been used here to create an image that represents the colours of the asteroid as perceived by the human eye. Credit: NASA/JPL-Caltech/UCLA/MPS/DLR/IDA

Preface

This is the 19th part of my Python tutorial series "Space Science with Python". All codes that are shown here are uploaded on GitHub. Enjoy!

Introduction

Today, we will cover another brightness related topic: the computation and determination of the apparent magnitude of asteroids. We cover some more

conceptual topics in the next article and then we will start with an asteroid related science project, coming with tutorial #20! If you need some information about the magnitude scale in astronomy, I would recommend to go first with my previous article:

Space Science with Python — Bright Dots in the Dark Sky

Part 16 of the tutorial series describes another important basic concept in Space Science: the brightness of objects.

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The complex nature of reflections

Did you take a look in the mirror this morning? Probably. But this space science article is not about your perception of yourself; it is about the laws of physics behind it. Plane mirrors and their reflective properties can easily be explained: ***Entrance Angle of the light*** = ***Exit Angle of the light***. Done.

But in space we do not have mirrors or simple geometries that allow us to compute the brightness of e.g., comets, meteors or asteroids. Most equations in this regard are empirically determined and are only valid within a certain solution space. In this article, we will continue with the topic that started some sessions ago: Asteroids.

Doing literature work means doing science

If you did not work in academia let me tell you one thing: every scientific project starts with literature work. *What did others do? Is there already a solution for a certain complex "niche" problem?* Where would you start to search for an *"equation or formula that is being used to compute the brightness of asteroids"*?

This task can be challenging since tons of papers, conference proceedings, articles and books are being released every day. Further, if you do not work at the university of any research facility, you will not have access to certain journals. However, most researchers upload their papers on a pre-print server named *arXiv* (the *X* is pronounced like the Greek "chi"):

arXiv.org

arXiv is a free distribution service and an open-access archive for 1,732,878 scholarly articles in the fields of...

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Further, scientists try to improve the work of others. *But are improvements correctly reviewed? Or are older empirical solutions sometimes better?* In our case, we search for the so-called *H-G magnitude/phase function* defined by Bowell et al. (1989). This is an empirical function to compute the brightness of asteroids that are still being used today!

Even newer versions, e.g., Muinonen et al. (2010), are less commonly known than the older H-G function [1].

So where do we find the corresponding equation? Luckily, since the H-G function is commonly used it can be found in free accessible papers and articles, like here:

Summary of the H-G magnitude system / function

So let's deep dive into the equation by extracting the relevant information for us:

Deep Dive

The H-G function provides the apparent/visual brightness V of an asteroid in magnitudes and it depends on:

- The distance between the Asteroid and the Sun
- The distance between the Asteroid and the observer (in our case: Earth)
- The size of the Asteroid, respectively the so-called absolute magnitude (similar as described in an article about comets)

Space Science with Python — Did we observe everything?

Part 11 of the tutorial series is about a Data Scientist's and Space Scientist's nightmare: Statistical Bias!

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- The phase angle between the Sun and Earth as seen from the Asteroid (so the enclosed angle between the directional vectors to the Sun and Earth with the Asteroid as an observer)
- The so called slope parameter G (we will describe this parameter in a few moments)

Let's take a look at the equation. As you can see, the distances to the Sun and Earth are embedded within a log10 function. Before, we have a so called reduced magnitude function $H(\alpha)$, with α being the mentioned phase angle. In fact, the function requires also the absolute magnitude H and the slope parameter G.

$$V = H(\alpha) + 5\log_{10}\left(d_{\text{ast,sun}} \cdot d_{\text{ast,ear}}\right)$$

Apparent / Visual magnitude of an asteroid

Let's write this equation in Python and let's also provide a proper doc-string at the beginning. Today, we will only need the packages numpy and matplotlib. The reduced magnitude is another function that is called in line 29 (Part 2/6).

Part 1/6

Part 2/6

So how does the phase angle depending reduced magnitude look like?

$$H(\alpha) = H - 2.5\log_{10}\left((1-G)\,\phi_1(\alpha) + G\phi_2(\alpha)\right)$$

Reduced magnitude

The reduced magnitude depends on the absolute magnitude H and other functions that are embedded within log10. ϕ_1 and ϕ_2 are called phase functions and are scaled

with the mentioned slope parameter G. Both functions have the same structure, but different coefficients. But first, let's code also this part in a second Python function:

Part 3/6

Again, the phase functions are similar, and have only different coefficients:

$$\phi_1 = \exp\left(-3.33\tan^{0.63}\left(\frac{1}{2}\alpha\right)\right)$$
$$\phi_2 = \exp\left(+1.87\tan^{1.22}\left(\frac{1}{2}\alpha\right)\right)$$

Phase functions 1 and 2

We can now code both phase functions in one Python function. The coefficients are stored in dictionaries (line 22 to 23 and 25 to 26). The input parameter *index* is a string that corresponds to the index number as shown in the equation above.

Part 4/6

To keep things simple, let's compute one example with certain parameters to visualise, analyse and interpret a particular brightness curve. Instead of using SPICE kernels, let's set some hard coded parameters that correspond with the properties of the dwarf planet Ceres:

JPL Small-Body Database Browser

Edit description

ssd.jpl.nasa.gov

Line 4 and 5 define a fixed distance between the object and the Sun (3.0 AU) and our home planet to the Sun (1.0 AU). Line 8 sets an absolute magnitude of 3.4 and the slope parameter *G* is set in line 9.

What do we want to achieve? Well, since almost all parameters are fixed, we will plot the apparent magnitude vs. the phase angle. Here, the required phase angle is the angle as seen from the asteroid. First, let's set an array with phase angles between the Earth and the asteroid as seen from the Sun (line 13). Now, we can apply trigonometric functions, the compute the distance between the asteroid and Earth, depending on the angle between Earth and asteroid as seen from the Sun (line 17 to 19). Our home planet is fixed and we move the asteroid from 0 degrees to 45 degrees in 1000 steps. Using these results, we can compute the phase angle of the asteroid in line 23 to 24. Finally, in line 27 to 31, we call our apparent magnitude function and compute the phase angle corresponding brightness values.

Part 5/6

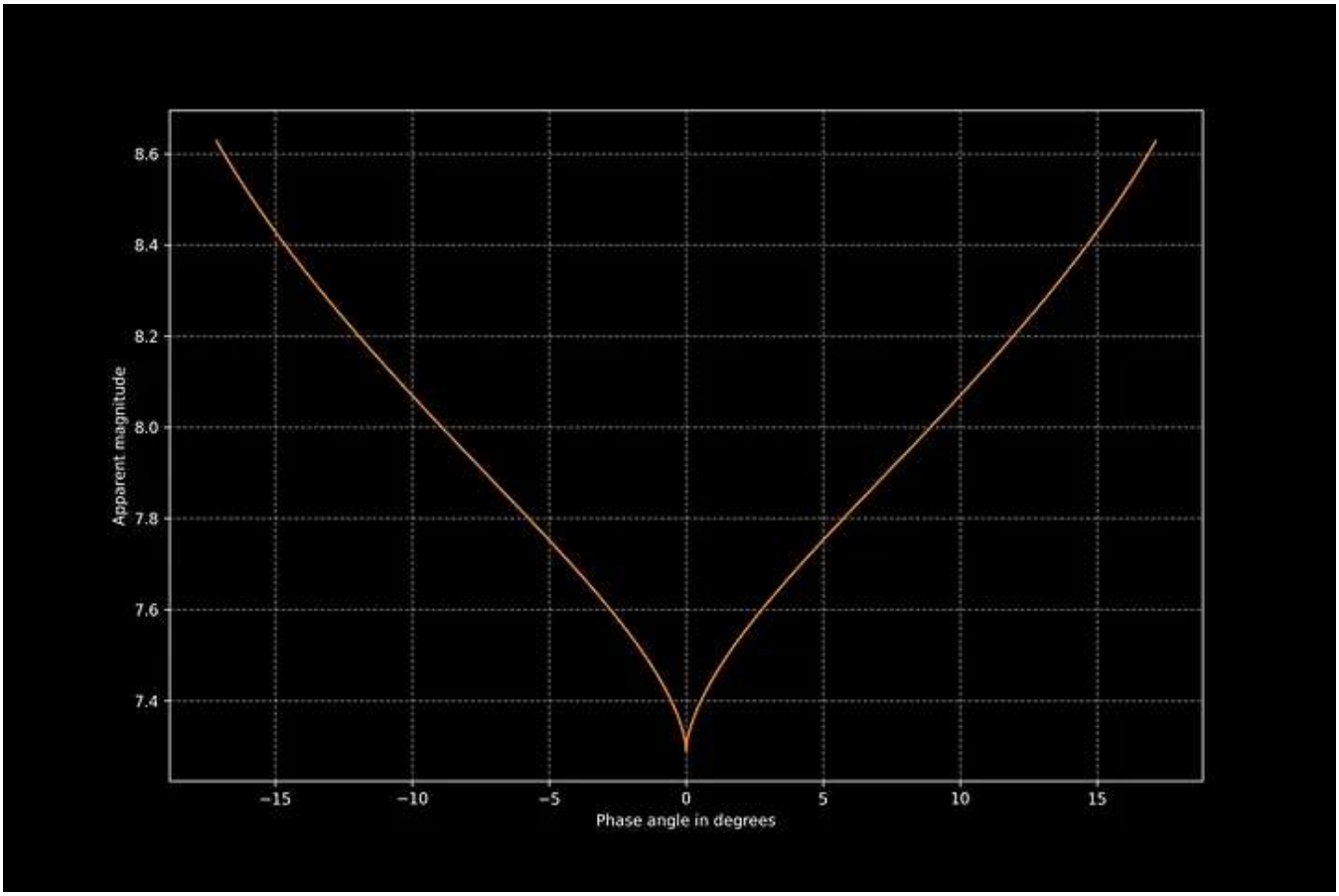
Now we can plot the results using matplotlib. We plot the magnitude vs. the phase angle as seen from the asteroid. For a better visualisation, we mirror the phase angles at 0 degrees. The resulting graph will become clearer (line 11 to 14). As already known from the previous tutorial session, we apply some formatting and label setting at the end.

Part 6/6

The resulting plot is shown below. You can see the apparent magnitude vs. the phase angle as seen from the asteroid in degrees. Please remember: a smaller magnitude value corresponds to a brighter object!

Between 5 degrees and 15 degrees the magnitude changes almost linearly with the phase angle. The reason is quite obvious: a larger phase angle corresponds to a smaller illuminated area of the object as seen from the Earth, similar as the waning and waxing moon phases. However, between 0 and 5 degrees the brightness curve becomes steeper and the apparent magnitude does not follow a simple linear model.

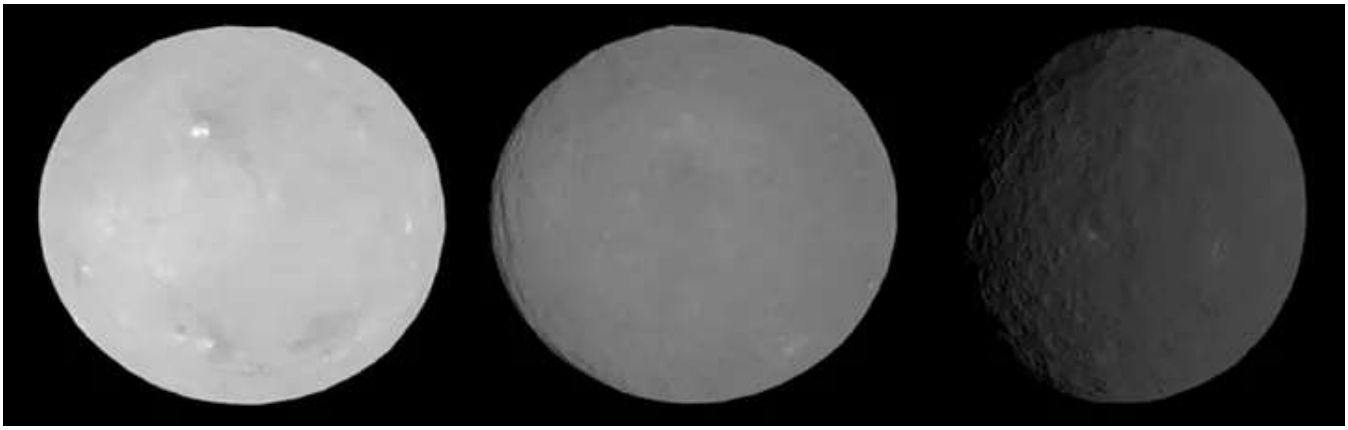
Is this an actual effect, or some modelling error, due to the empirical nature of the equation?



Apparent magnitude vs. phase angle in degrees. Negative phase angles are mathematically not defined. Here, the curve on the right hand side has been mirrored to improve the opposition effect / surge. Credit: T. Albin

Opposition Effect

What you see is the so called ***Opposition Effect*** or ***Opposition Surge***. The brightness increases dramatically for very small phase angles and was also well observed and described by the Dawn mission that orbited the dwarf planet Ceres [2].



Apparent brightness of Ceres as seen from Dawn. Phase angles from left to right: 0°, 7° and 33°. Data from Wikipedia; Credit: NASA/JPL-Caltech/UCLA/MPS/DLR/IDA

The same effect can also be observed in our daily lives! Go to the beach and stand on the sand with the Sun in your back. Look down to the shadow of your head and you will see a bright halo around your head's shadow!

How is this effect explainable? The reason for this effect results from the asteroid's surface properties (or the sand beach). The top layer is not smooth but has a layer of dust, small rocks, pebbles … so called ***regolith***. This regolith particles cause a shadow when light shines down to the surface. However, the regolith particles are throwing shadows depending on the illumination angle. Like a tree, whose shadow is way longer during the evening hours (large phase angles) than during noon. Small phase angles, respectively a perpendicular illumination of the surface reduces the "self-shadowing" abruptly! The effective illuminated area increases and causes a sudden brightness surge for small phase angles.

The shape of this surge between 5 and 0 degrees is defined by the slope parameter G. It is assumed that most asteroids have a value at around 0.15.

Conclusion

It is up to you now. Get familiar with the H-G function and change the parameters. Check how the opposition surge changes for G=0.1 or G=0.5. Try to create an animation where you dynamically change several settings …

Space Science with Python — Supplements for Papers

Part 10 of the tutorial series is a Supplementary Article on Supplementary Materials. How can we visualise…

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… or try to use actual orbit data of asteroids:

Space Science with Python — Uncertain Movements of an Asteroid

Part 17 of the tutorial series. Today (30th June) is Asteroid Day and we will start to deep dive into the science of…

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In the last couple of weeks you learned several methods, tools and algorithms. Combine these things to generate new insights! The mixture of examples and basics / fundamental concepts should allow you to become a more and more independent citizen space scientist.

Next time we will combine the H-G function with actual data and use sky maps. Afterwards, we will start with our asteroid science project! I hope you enjoyed today's lesson.

Thomas

References

[1] Karri Muinonen, Irina N. Belskaya, Alberto Cellino, Marco Delbò, Anny Chantal Levasseur-Regourd, et al.. A three-parameter magnitude phase function for asteroids. Icarus, Elsevier, 2010, 209 (2),pp.542–555. 10.1016/j.icarus.2010.04.003. https://hal.archives-ouvertes.fr/hal-00676207/document

[2] Schröder, Stefan E.; Li, Jian-Yang; Rayman, Marc D.; Joy, Steven P.; Polanskey, Carol A.; Carsenty, Uri; Castillo-Rogez, Julie C.; Ciarniello, Mauro; Jaumann, Ralf; Longobardo, Andrea; McFadden, Lucy A.; Mottola, Stefano; Sykes, Mark; Raymond, Carol A.; Russell, Christopher T.; Ceres' opposition effect observed by the Dawn framing camera; Astronomy & Astrophysics, Volume 620, id.A201, 19 pp.; 2018; DOI: 10.1051/0004–6361/201833596

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